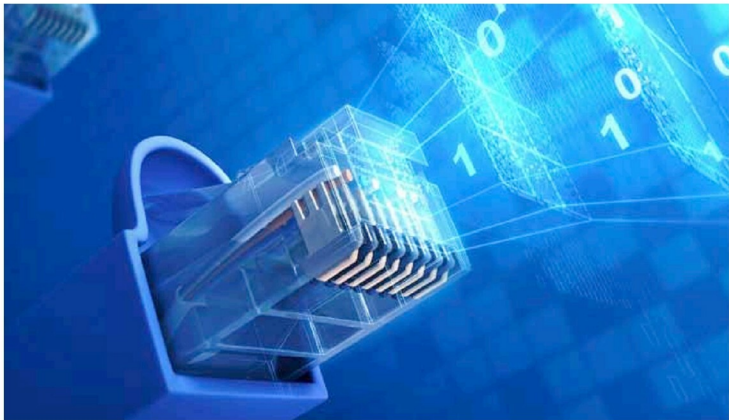


Analyse Your Network Packets with LibPCAP

Data on a network flows in the form of packets, which are entities that carry your data across the network. These packets are routed from one router to another by intelligent nodes. LibPCAP is a library that lets you 'capture' these packets.



Packet capture lets you intercept and entirely grab any packet (with all its headers) that is seen by the network device. Packet capture is done regardless of which port the packet is sent to, or even regardless of the host. LibPCAP provides an implementation independent library that lets you access the underlying packet capture facility provided by your operating system. Hence, libPCAP is used to capture packets, directly off the network cards.

Pre-requisites

There are a few things you should know before you can implement the code provided in this article.

Here is a list of things you might want to review before proceeding to read this article.

1. Basics of C programming
2. The basics of networking and its terminology

Getting started

The only thing you need to set up, to run the codes provided

in this article, is a working C/C++ compiler (I have used gcc in this article) and libPCAP. To install libPCAP on Ubuntu, you can simply run the following command in a terminal:

```
apt-get install libpcap-dev
```

In other distributions, you can download the source package for libPCAP from <http://www.tcpdump.org/#latest-release> and build it with the following commands:

```
./configure  
make  
make install
```

Now that we have set up everything we need, let us try and understand libPCAP better.

Understanding libPCAP

Let us have a look at the structure of the PCAP packet sniffer